

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm:

exactly 80 mm: _____

4 YELLOW CROSS

yellow edges only – ignore the corners



Dot no yellow edge at all
FRU'R'U'



Hook two of a right angle; hold them BACK and LEFT
FRU'R'U'



Line hold the bar left-to-right
FRU'R'U'

One algorithm, up to three times; dot to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn it so at least two top edges match the center under this. This algorithm is called *Sune*; you use it on every card after this one.



two matched, side by side hold them BACK and LEFT
FRU'R'U'



two matched, opposite run it once from any hold, then look again
FRU'R'U'

6 PUT THE YELLOW CORNERS HOME



one corner home hold it FRONT-LEFT
RU'L'U' RU'L'



no corner home run it once from any hold, then look again
RU'L'U' RU'L'

This cycles the other three corners into place. It also twists the yellow face you just look expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS



yellow faces RIGHT
R'D'R'D' x2



yellow faces FRONT
D'R'D'R' x2

Between corners run **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only card on this card that gets reset.

UNDOCK CARD 2 – five scramble in a row with this card face down. **0 MASTER:** those five under 300. My target: _____ NEXT: 2/3 TWO-LOCK

YELLOW CROSS yellow edges only – ignore the corners

 Dot no yellow edge at all
FRUR'U'F'

 Hook two of a right angle: hold them BACK and LEFT
FRUR'U'

 Line hold the bar left-to-right
FRUR'U'F'

One algorithm, up to three times: dot to hook to line to cross.

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First turn it so at least two top edges match the center under this. This algorithm is called *Sune*; you use it on every card after this one.

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Between corners run **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

LAST-LAYER ORDER on THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only thing on this card that gets retired.

UNDOCK CARD 2 – five scramble in a row with this card face down. o MASTER: those five under 300. My target: _____ NEXT: 2/3 TWO-LOCK



PLL CORNERS – headlights = 2 matching corners on one face

T headlights on ONE face – hold that face LEFT
RUR'U' R' R2UR'U' RUR'
 T-Perm. No headlights to hold left? Run it twice.

Y no headlights – the two swapping corners sit diagonally
FRU'R' RUR'F RUR'U' R'FR'
 Y-Perm. No headlights to hold left? Run it twice.



PLL EDGES – corners home, top still not solved

UB the front edge travels LEFT
R2U RUR'U' R'U' RUR'
 UB-Perm. No headlights to hold left? Run it twice.

UR the front edge travels RIGHT
M2U M2U' M2UM2
 UR-Perm. No headlights to hold left? Run it twice.



H both pairs swap straight across
M2UM2 U2 M2UM2
 H-Perm. No headlights to hold left? Run it twice.



Z each pair swaps with its neighbour
M'U' M2UM2M'U' M'U2 M2U
 Z-Perm. No headlights to hold left? Run it twice.

THE NEW ORDER: yellow, yellow, face, corners, home, edges. Home never changes again. Card 1's order is retired, and so is Niklas; it moves corners but wrecks the yellow face, and here the yellow face is finished first.

UNCLACK CARD 3 – different last layers in a row, each finished in exactly two solves averaging under 1:00. My target: 1:00.00.

NEXT: 3-ONE LOOK PLL

P1L CORNERS = headights = two matching corners on one face



T headlights on **ONE** face – hold that face LEFT
RUR'U' R' R' R2UR'U' RUR'



Y no headlights – the two swapping corners sit diagonally
FRU'R' RUR' R' RUR' R' FRF'

START HERE – two new algorithms finish any last layer
 yellow cross **F R U R' U'** until the cross appears – at most 3.
 Yellow face: Sube, read again, repeat – at most 3 (see the badges).
 Corner: T-Perm. No headlights to hold at back! Run it twice.
 Edges: U. No solved edge to put at the back! Run it twice.
 Each case on the front replaces one repeat with one turn. Three a week, you can solve the whole lot all four faces.

HS Z
 Both show headlights on all four faces. The edge between them decides if it belongs to the opposite face = R, to a neighbour = Z.
STUCK?
CORNERS hurt but nothing solves? Turn the top face once and then read again – that is often the whole lot.
 One case has beaten you five times! Park it, use its fallback above, can back next week.
 A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and reseat it.

P1L EDGES = corners home, top still not solved

UB the front edge travels LEFT
R2U RUR'U' R' R' UR'

UR the front edge travels RIGHT
M2U M2U M'U M2U

WORDS
OLL = make the whole top face yellow - P1L = slide the top pieces home - headights = two matching corners on one face - AUF = the free top that lines a case up

H both pairs swap straight across
M2UM2 U2 M2UM2



Z each pair swaps with its neighbour
M'U' M2UM2 M2U' M'U2 M2U

THE NEW ORDER: yellow cross, yellow face, corners home, edges home. This never changes again. Card 1's order is right, and so is Niklas: it moves corners but wrecks the yellow face, and so the next face is finished first (UNLACK CARD 3 – different last layers in a row, each finished in exactly two turns).

solves averaging under 1:30. My target is **1:00** – NEXT: 3x ONE-LOOK PLL

EDGES ONLY corners already home • Ua B solid: F.L.R headlights: 3 edges counter-clockwise M2U MU2 M'UM2  • Ub B solid: F.L.R headlights: 3 edges clockwise R2UR' R'U' R'UR'  • H all 4 headlights: opposite edges swap M2U'M2 U2 M2U'M2  • Z all 4 headlights: adjacent edges swap M2U'M2MU2U' MU2 M2U 	DIAGONAL CORNER SWAP – no headlights anywhere  • Y pairs F-left R-back F'RUR'U' RUR'F' URUR' R'RF'  • Y pairs F-left L-front r'Yr'U' r'Yf' R2UR'U R'FF'  • No pairs R-left F-right L-back R-back R'UR'U'R' FR'U'R' RUR'F' R'F' R'UR' UZUR'  • Nb pairs R-right F-left L-back R-front R' URUR' F'UF' RUR' FR'F' R'UR'  • X' nothing matches E L'ULD' L'ULD' L'ULD' L'ULD
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EDGES ONLY – corners already home

- **Ua** B solid: F.L.R headlights: 3 edges counter-clockwise
M2U MU2 M'UM2
- **Ub** B solid: F.L.R headlights: 3 edges clockwise
R2U R'UR' R'U' R'UR'
- **H** all 4 headlights: opposite edges swap
M2U'M2 U2 M2U'M2
- **Z** all 4 headlights: adjacent edges swap
M2U'M2MU2U' MU2 M2U

DIAGONAL CORNER SWAP – no headlights anywhere

- **Y** pairs F-left R-back
FRU'R'U' R'UR'F' R'UR'U' R'RF'
- **Y** pairs R-left L-front
R'UR'U' Y R'F' R2UR'U' R'RF'
- **Na** pairs R-left F-right L-back
RU'R'U' R'UR'F' R'UR'U' R'F'
R'UR'U' U2RU'R'
- **Nb** pairs R-right F-left L-back R-front
R' URUR' F'U'F RUR' FR'F'
R'UR'
- **E** nothing matches
X' L'UL'D' L'UL'D L'UL'D'
L'UL'D

HOW TO LOOK

- Count the height/faces. One = front of this card. Four = this column. None = the column on the right.
- Only then read the edges to pick the exact case.
- Line the case up as drawn, run it, turn the top again to finish – that last free turn is the AUF.
- Always starts from Card 2.

WORDS

PLS = slide the top pieces home - headlights = 2 matching corners on top - AUF = free top turn that gives a case up adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners are opposite

Card 2 gives you 19/72 last layers in one look. This card gives 71/72. Next: the other 37 face-matched cases, the first two layers solved as pairs – drilled with a cube in the app: cube3step3s vereel.app/practice

DONE – all twenty-seven named correctly, twenty-one in a row, no separate days, on two separate days, in 10 minutes, under 630. My target: 10 minutes.

THE LADDER

1/3 FIRST LOOK - 5 algorithms - unlock: five solves, card face
 2/3 TWO-LOOK - 11 unlocks: fifteen last letters in exactly two
 3/3 ONE-LOOK PL1 - +15: done! all 21 named, twice on separate
 days

So that: full OLL, F2L, cross planning, look-ahead – in the app
 not on card stock.

WHY THERE ARE ONLY THREE

1. The first two are the only ones you tell apart by sight. It is
 bad at this you drill F2L, cross planning and look-ahead are drills
 with no finite case tell. Full OLL is 57 cases, 22 of which differ only
 by a diver 4 fifth of a millimetre wide at this size. All of them are
 in the app, with a real cube and randomised setup. This set stops
 where the medium stops.

PRINT AND VERSION

EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were
 to wonder the build would follow the same path. Recognition wording is generated from the same permutation diagram is drawn from.

Set V1.0: **reprint** any single card at cubepu3h-s1s.vercelapp/print - print at 100% / Actual size, never Fit to page!

Not a tier, no unlock, no record. Use it when you buy a 4.

THE LADDER

1/3 FIRST LOOK - 9 algorithms - unlock: five solves, card face
 2/3 TWO-LOOK - 11 + unlock: fifteen last letters in exactly two
 3/3 ONE-LOOK - 115 - add: all 21 named, twice on separate
 days

Not a card that: full OLL, Drill, cross planning, look-ahead - in the app not on card stock

WHY THERE ARE ONLY THREE

1/3 FIRST LOOK - 9 algorithms - unlock: five solves, card face
 2/3 TWO-LOOK - 11 + unlock: fifteen last letters in exactly two
 3/3 ONE-LOOK - 115 - add: all 21 named, twice on separate
 days

Not a card that: full OLL, Drill, cross planning, look-ahead are drills with no finite case table. Full OLL is 57 cases, 22 of which differ only by a diver's fifth of a millimetre wide at this size. The other two are in the app, with a real cube and randomised setup. This sets steps where the medium step

PRINT AND VERSION

EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if any were wrong the build would fail and the program would be rejected. The diagram is drawn from
 Set V1.0 - replace any single card at cubepu3h15s.vercelapp/print - print at 100% / Actual size, never Fit to page'

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